

ANUJ YADAV

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EDUCATION

University of Delhi

Bachelor of Science (Hons.) in Electronics

Score: 7.63 CGPA (Till 5th Semester) | Keshav Mahavidyalaya

Delhi, India

Expected May 2026

Sarvodaya Vidyalaya Bhalswa Village (Delhi)

AISSCE (Science, Class XII)

Score: 79.6% [398/500]

Delhi, India

2023

Sarvodaya Vidyalaya Bhalswa Village (Delhi)

AISSE (Class X)

Score: 86.6% [433/500]

Delhi, India

2021

RELEVANT COURSEWORK

- Data Structures • Algorithms Analysis • Database Management • Operating Systems
- Digital Electronics • Embedded Systems • Communication Systems • Microprocessors (8085)

PROJECTS

Smart Home Automation System | Embedded C, ATmega328P, GSM, L293D | [Live Demo](#) ■

2025

- Automated home subsystems (fan speed, light, fire alarm, windows) via MQ-3, PIR, LM35 and rain sensors driven by ATmega328P.
- Integrated SIM900 GSM module over bare-metal UART using AT commands to send automatic emergency warning texts under 3 seconds.
- Published the circuit online as a visual firmware execution system simulating digital registers and pin loops live in-browser.

Checkmate — Chess Engine with AI | Minimax, Alpha-Beta Pruning, Heuristics | [Live Demo](#) ■

2026

- Programmed logical verification structures supporting castling, pawn promotions, legal paths, checkmate and stalemate flags.
- Engineered localized evaluation heuristic maps optimized with minimax algorithm trees and alpha-beta pruning pathways.

VoltLab — Logic Circuit Simulator | Canvas Engine, Graph Simulation, Fixpoint Solver | [Live Demo](#) ■

2026

- Created graphical wiring simulation modeling logic gates (AND/OR/XOR) to switch inputs and live LED outputs on canvas.
- Programmed cycle loops & fixpoint iteration feedback resolvers alongside automatic 4-input dynamic truth-table solvers.

Gridlock — Spreadsheet Formula Engine | Recursive-Descent Parser, Dependency Graph | [Live Demo](#) ■

2026

- Wrote recursive-descent math formula compilers calculating cell dependency values (SUM, AVG, IF) automatically.
- Engineered circular graph solver complete with Excel key hotkeys, formula statusbars and fast CSV import/export options.

Codeplay — Live Code Playground | Sandboxed HTML iFrames, postMessage API | [Live Demo](#) ■

2026

- Constructed front-end compiling engine rendering script snippets live inside secure isolated sandboxed iframe modules.
- Established custom postMessage bridges capturing runtime error indicators and console reports dynamically.

FocusForge — Pomodoro & Sound Studio | Web Audio API, Audio Oscillators, LocalStorage | [Live Demo](#) ■

2026

- Designed a focus app with audio oscillators, brown-noise synthesizers, weekly task queues and localStorage trackers.

AI Resume Analyzer | Firebase, Analytics, Document Parsing | [Live Demo](#) ■

2026

- Shipped client-side ATS analysis tool tracking word counts, keywords, and recommending profile optimization steps.

AVR Embedded Bare-Metal Practicals | Embedded C, ATmega32, Timers & Keypads

2025

- Programmed 15+ low-level bare-metal hardware routines managing timers, interrupts, and registers in Proteus and Wokwi.

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, HTML/CSS, 8085 Assembly

Embedded: Arduino UNO, ATmega32 (bare-metal AVR), UART/AT commands, L293D, sensor interfacing (LM35, PIR, MQ-3, rain), 16x2 LCD, timers & interrupts

Developer Tools: Git/GitHub, Proteus 8, Wokwi, MATLAB, Scilab, Google Colab, Linux (Termux), VS Code

Concepts: Data structures & algorithms, recursive-descent parsing, game-tree search, DBMS, operating systems